

APEX LABS

TEAM H



*Automated Precision EXperimentation
Laboratories*





PROGRESS REVIEW 4

CURRENT PROGRESS & CHALLENGES

PERCEPTION

Goal

- Complete fine-tuning models for detection and segmentation.
- Generate point clouds for planning.

Achieved

- All goals achieved.

Challenges

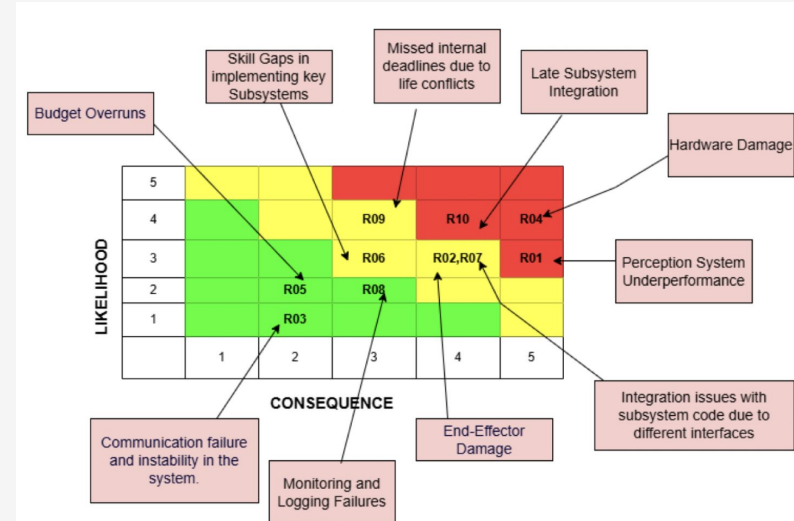
- None

PERCEPTION - ISSUES LOG

Item Number	Date Initiated	Date Resolved	Participants	Description	Options	Resolution	Justification
4.	03/01	03/15	Farnaz, Arnav	Capture Card not compatible with Jetson Thor	Do camera processing on Xavier	Doing camera processing on Jetson Xavier and streaming out to Thor	We need the component for streaming real-time. \

PERCEPTION - RISK

R01	Perception System Underperformance	Technical	3	5	1. Collect diverse datasets early and fine-tune detection models. 2. Add Aruco markers to aid in our detection.
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PLANNING

Goal

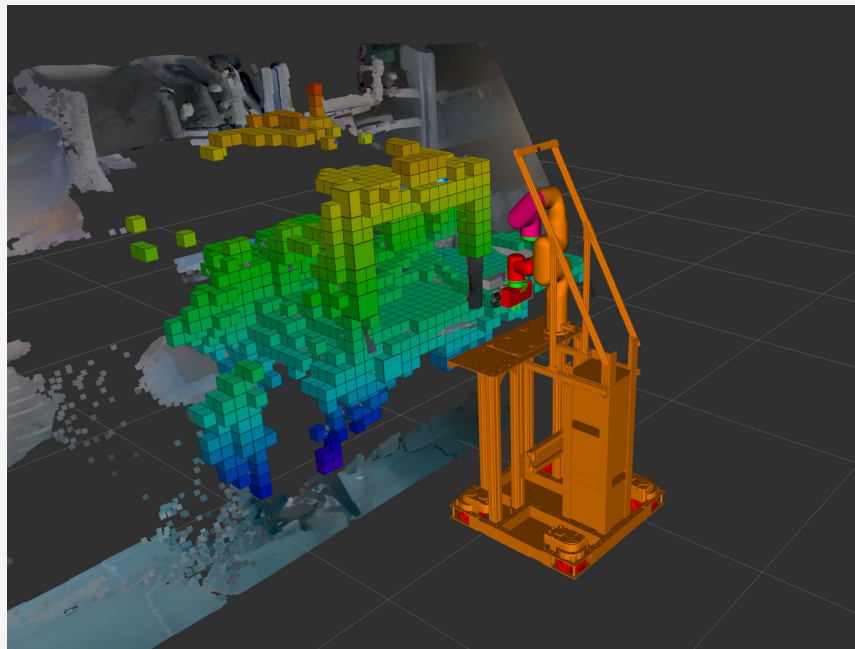
- Complete integration with manipulation (gripper)
- Begin integration with perception
- Test case by case for SVD workflow

Achieved

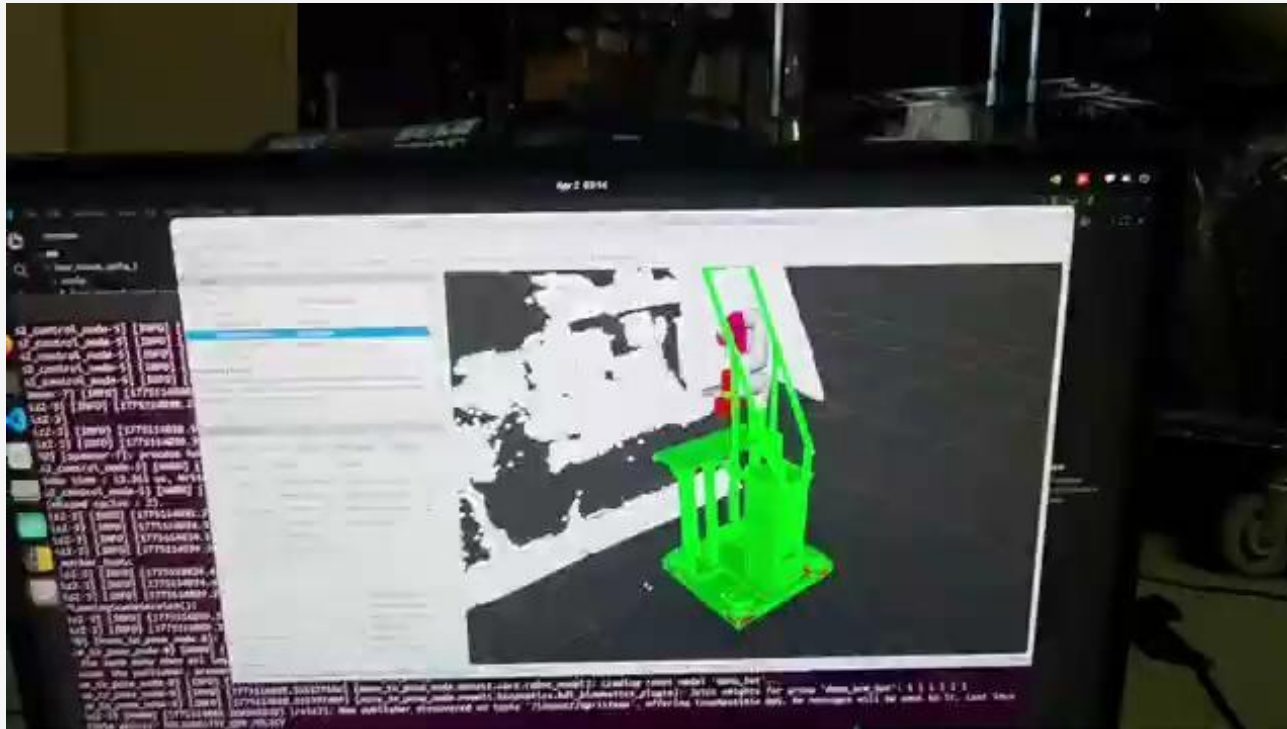
- Loaded point cloud from the live camera feed, began integration with perception
- Path planning to a location with AprilTag (OT2 slot)

Challenges

- Arm sometimes comes in LoS for the top cam
- Gripper broke again
- Shorting happened in an old Jetson Xavier
- Point cloud sometimes considered arm as collision object



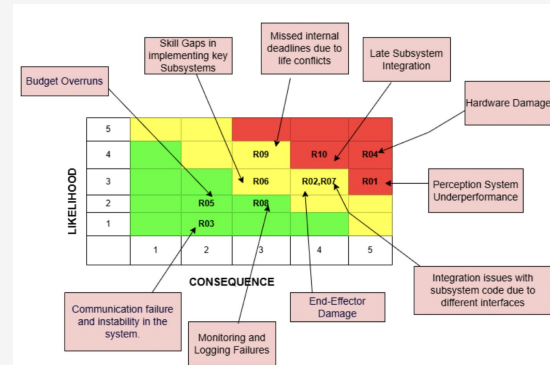
Point cloud visualization in RViz, along with the voxel map used for obstacle avoidance



Rviz -> Real robot

PLANNING - RISK

R07	Integration issues with subsystem code due to different interfaces	Technical; Schedule	3	4	1.Enforce strict version control practices using systems like Git. 2.Implement continuous integration (CI) practices to test the integration of subsystems frequently. 3.Clearly define API interfaces between subsystems, ensuring compatibility in terms of data types, formats, and protocols.
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MANIPULATION

Goal

- Test Visual Servoing with gripper.
- Can command the xArm6 to grasp well plates and transfer them.

Achieved

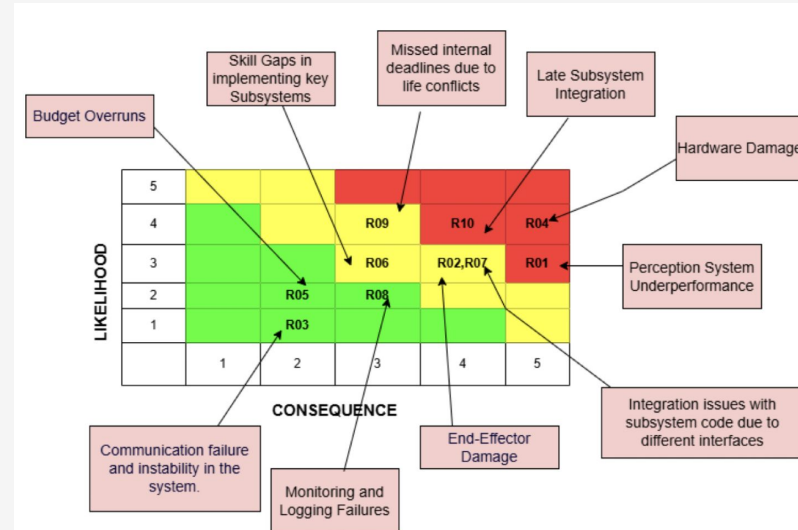
- All Achieved

Challenges

- Wrist Camera occlusion with end effector
- ZED X Camera error is quite frequent

MANIPULATION - RISK

R08	Monitoring and Logging Failures	Technical	2	3	1.Stress-test streaming pipelines under continuous load. 2.Use Filebeat with persistent queue configurations. 3.Store local MCAP ring buffer backups independent of the cloud logging stack.
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MISCELLANEOUS

Goal

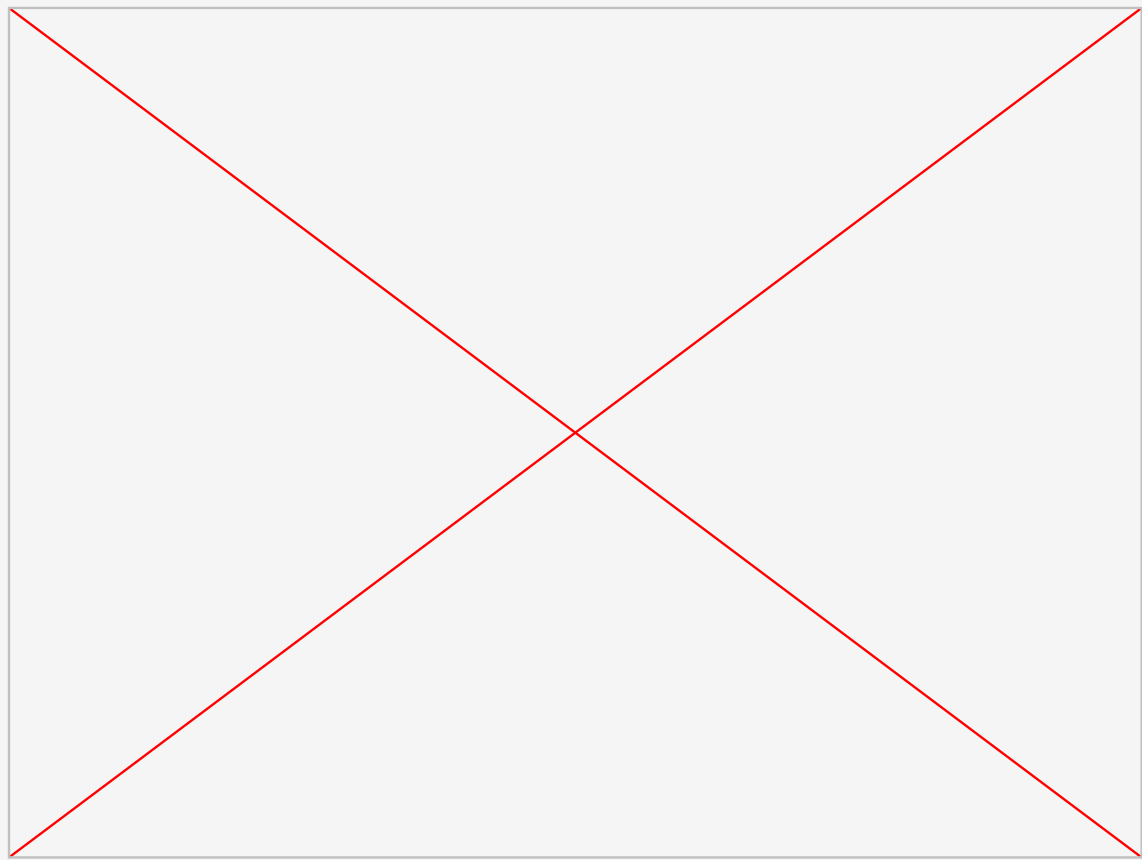
- Able to see the camera streaming from the UI.
- Integrate LMI subsystem with Behavior Tree.

Achieved

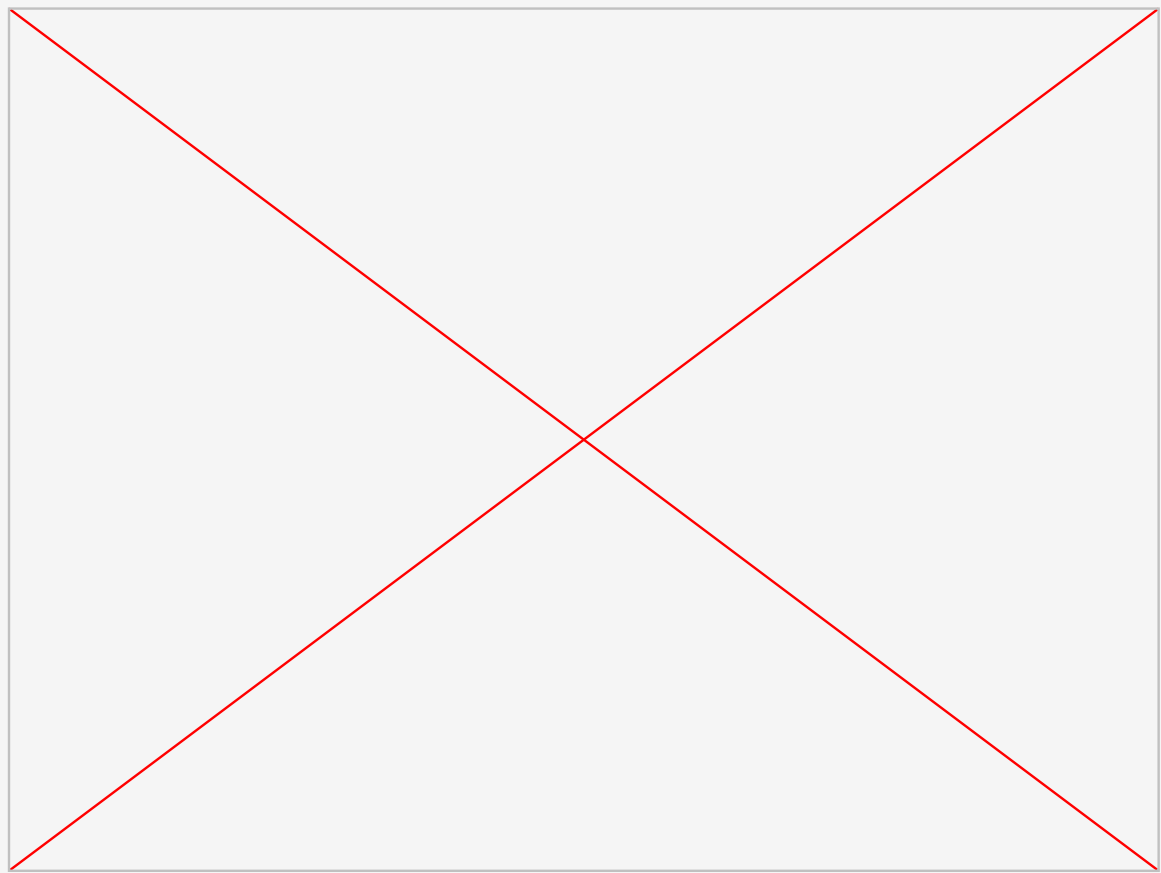
- Able to stream the camera to the UI.
- Integrated LMI subsystem with Behavior Tree.

Challenges

- Sharing the cameras amongst team members (since we can only open a camera in a single process).
- New approach is to create a single application that multi-threads all of the camera access points.



Camera Stream integrated into UI.



Test Tube integration and OT2 operating from Behavior Tree (autonomy)

FUTURE WORK

PERCEPTION

- Test out models and more locations across the campus and fine-tune the models.

PLANNING

- Complete integration with perception and manipulation
 - Ensure point cloud is filtered properly
 - Once object detection outputs 3d coordinates, I will use that to verify perception + planning stack
 - Visual servoing is done. So I can immediately test planning + manipulation integration as the next step.
- Integrate with behavior tree
- Tune as much as possible and conduct dry runs for SVD

MANIPULATION

- Integrate visual servoing with behaviour tree

MISCELLANEOUS

- Centralize camera access code
- Populate Shaker PCBv3 (DevKit)
- Designed mounting platforms for Xavier.
- Designing mounting platforms for emergency switch and USB Port.

QUESTIONS?

